Specifications

Ford Ecat Electronic Parts Catalog

Material

Item	Specification	Fill Capacity
Motorcraft® Bug and Tar Remover ZC-42	—	—
Motorcraft® Clear Silicone Rubber TA-32	ESB-M4G92-A	—
Motorcraft® Metal Bonding Adhesive TA-1	—	—
Motorcraft® Acid Neutralizer ZC-1-A	—	—
Motorcraft® Alkaline Neutralizer ZC-2-A	—	—
3M™ Perfect-It™ Show Car Liquid Wax 39026	—	—
Motorcraft® Detail Wash ZC-3-A	ESR-M14P4-A	—
Motorcraft® Metal Surface Prep ZC-31-B	—	—
Premium Undercoating ValuGard™ VG101, VG101A (aerosol)	—	—
Rust Inhibitor ValuGard™ VG104, VG104A (aerosol)	—	—
Motorcraft® Plastic Bonding Adhesive TA-9	—	—
Roof Ditch Sealer Fusor® 122EZ, 3M™ 08307	—	—
Motorcraft® Seam Sealer TA-2	—	—
Trim and Weatherstrip Adhesive Permatex® Black Super Weatherstrip Adhesive, 81850	—	—
Motorcraft® Ultra-Clear Spray Glass Cleaner ZC-23	ESR-M14P5-A	—

General Equipment
3 Phase Inverter Spot Welder 254-00002
Compuspot 700F Welder 190-50080
I4 Inverter Spot Welder 254-00014
Inverter Welder with Metal Inert Gas (MIG) Welder 254-00015

General Specifications - Welding Specifications

Item	Specification
Plug Weld Hole	8 mm (0.31 in)
Weld Wire ER70S-3 or equivalent	0.9-0.11 mm (0.035-0.045 in)

Weld Nugget Chart

Test Thickness of Metal (mm)	Nugget Size
0.7 + 0.7	4.3 mm (0.16 in)
0.7 + 0.7 + 0.7	4.3 mm (0.16 in)
0.9 + 0.9	4.7 mm (0.18 in)
0.9 + 0.9 + 0.9	4.7 mm (0.18 in)
1.0 + 1.0	5.2 mm (0.2 in)
1.0 + 1.0 + 1.0	5.2 mm (0.2 in)
2.0 + 2.0	7.1 mm (0.27 in)
2.0 + 2.0 + 2.0	7.1 mm (0.27 in)
3.0 + 3.0	8.7 mm (0.34 in)
3.0 + 3.0 + 3.0	8.7 mm (0.34 in)
3.0 + 0.7	4.3 mm (0.16 in)
0.7 + 3.0 + 1.0	5.2 mm (0.2 in)
2.0 + 2.0 + 0.7	4.3 mm (0.16 in)
0.9 + 0.9 + 2.0	4.7 mm (0.18 in)
2.0 + 0.9 + 1.0	5.2 mm (0.2 in)
1.0 + 3.0 + 1.0	5.2 mm (0.2 in)
3.0 + 1.0 + 2.0	7.1 mm (0.27 in)
0.9 + 0.7 + 0.9	4.3 mm (0.16 in)

Descriptions of Ford Steel Families

Grade	Alloy Content	Heat Treatment	Typical Applications	Comments
Mild Steel, Bake Hardened, Solid Solution Strengthened	Low	Fully annealed/dead soft	Body panels (closures, floor pan, dash panel)	—
High-Strength Low Alloy (HSLA)	Low	Fully annealed/dead soft	Rails, structural members	Strengthened with fine particles and small grain size
Dual Phase Steel (DP)	Medium (Manganese Silicon, Molybdenum Chromium)	Fully annealed/partially hardened	Rails, structural members	15-50% of structure is hard martensite
Ultra High Strength Steel (UHSS) (Martensitic, Boron)	Low	Fully hardened	Rocker reinforcements, door beams, bumper beams	100% of structure is hard martensite
Transformation Induced Plasticity Steel (TRIP)	High (Manganese, Phosphorous, Silicon, Aluminum)	Fully annealed/partially hardened	To be determined	Complex microstructure for high strength and ductility

Ford Recommended Steel Repairability Matrix

Grade	Trade Descriptions	Welding Method			Cold Repairs	Use of Heat for Repair	Temperature Range	Maximum Heat
		Metal Inert Gas (MIG)	Squeeze-Type Resistance Spot Welding (STRW)	MIG Braze				
Mild Steel	Mild	Yes	Yes	N/A	Yes [a]	Yes	Up to 650°C (1,200°F)	90 sec. x 2
Laminate Steel	Quiet Steel	No	Yes	No	Yes[a]	N/A	N/A	N/A
Bake Hardened Steel (BH)	BH 180, BH 210, BH 250, BH 280	Yes	Yes	Yes [b]	Yes[a]	Yes	Up to 650°C (1,200°F)	90 sec. x 2
Solid Solution Strengthened	—	Yes	Yes	Yes[b]	Yes[a]	Yes	Up to 650° C (1,200°F)	90 sec. x 2
High-Strength Low Alloy (HSLA)	HSLA 250, HSLA 350, HSLA 550	Yes	Yes	Yes[b]	Yes[a]	Yes	Up to 650°C (1,200°F)	90 sec. x 2
Dual Phase Steel (DP)	DP 500 class, DP 600 class	Yes	Yes	Yes [b]	Yes[a]	No	N/A	N/A
DP [c]	DP 700 class, DP 900 class	Yes [d]	Yes	Yes[b]	No	No	N/A	N/A

Grade	Trade Descriptions	Welding Method			Cold Repairs	Use of Heat for Repair	Temperature Range	Maximum Heat
		Metal Inert Gas (MIG)	Squeeze-Type Resistance Spot Welding (STRW)	MIG Braze				
	and DP 1,000 class							
Ultra High Strength Steel (UHSS) (Martensitic, Boron) [e]	Boron	Yes[a]	Yes	Yes[b]	No	No	N/A	N/A
Transformation Induced Plasticity Steel (TRIP)	TRIP 590, TRIP 780, TRIP 980	N/A	N/A	N/A	N/A	N/A	N/A	N/A

[a]Cold repairs can be performed if damage excludes kinks. May section only if approved procedure in workshop manual.

[b] MIG braze allowed for non-structural applications only.

[c]Dual phase steels DP 700, DP 780, DP 980 and DP 1000 must be replaced at factory joints, no sectioning unless approved procedure in workshop manual.

[d]For DP 980 and DP 1,000 class, use Metal Inert Gas (MIG) plug welding only, no stitch welding.

[e]Boron components must be replaced at factory joints, no sectioning allowed.